Modeling the Effect of Lipid Degradation in Microalgae

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Introduction

Background

- Marine microalgae have been studied because of their high lipid production which can be used to generate biofuels.
- Mathematical models have been developed to understand the processes of algal growth and to *optimize conditions* for lipid production.

Previous Work

- Current models fail to predict lipid production under a wide array of stress conditions.
- Our model aims to use a single set of parameters to predict lipid accumulation under different nitrogen and light stresses.

Problem

- In our experimental results we observed degradation in the accumulated lipids.
- The model already created by our research group does not currently capture this event.

Objective

- **Understand the degradation** mechanism observed in experimental results and **improve the performance** of the mathematical model.

Results and Discussion

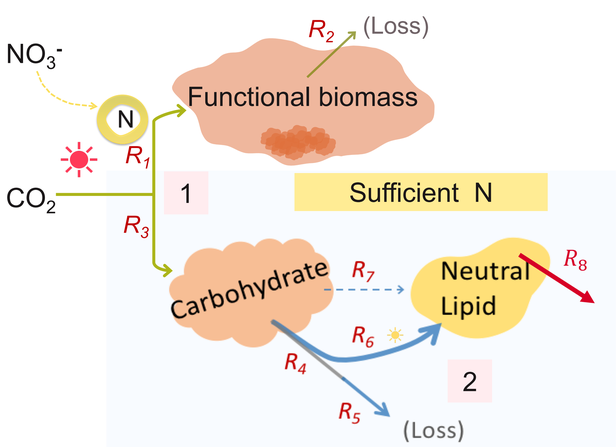
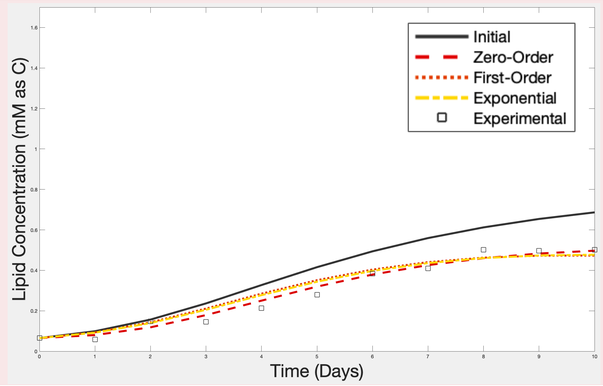


Figure 1: Schematic diagram of storage molecule production regulated by nitrogen and light.

Results and Discussion (cont.)

Calibration Set

Data Set 1: Low Light, High Nitrogen

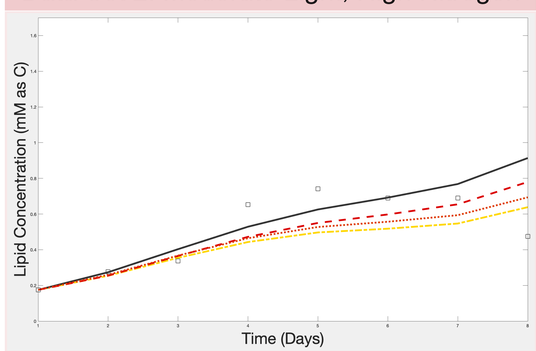


Modeling Equation	Parameter Values
Zero-Order Rate Reaction $= dX_{nl} - k_1$	$k_1 = 0.0188$
First-Order Rate Reaction $= dX_{nl} - k_2 \cdot X_{nl}$	$k_2 = 0.0679$
First-Order Exponential Decay: $= dX_{nl} - k_3 \cdot X_{nl} \cdot (1 - \mathrm{erf}(-k_4 \cdot I_{avg})$	$k_3 = 0.0468;\ k_4 = 0.1988$

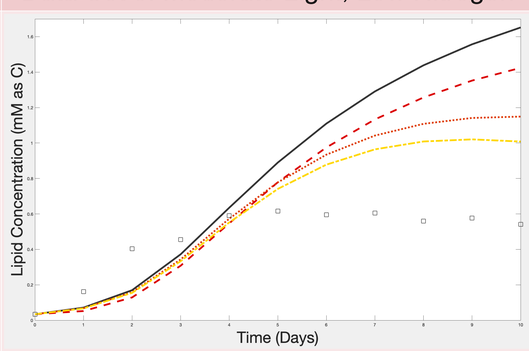
$$dX_{nl} = \beta \cdot \exp(k_{nl} \cdot I_{avg}) \cdot v_n \cdot X_{cd} + q_{nl} \cdot \frac{q_n}{Q_n} \cdot P \cdot X_{cd}$$

Validation Sets

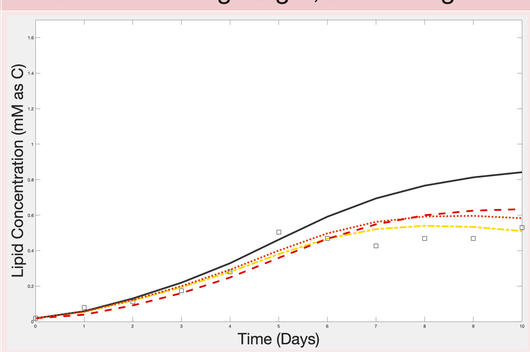
Data Set 2: Moderate Light, High Nitrogen

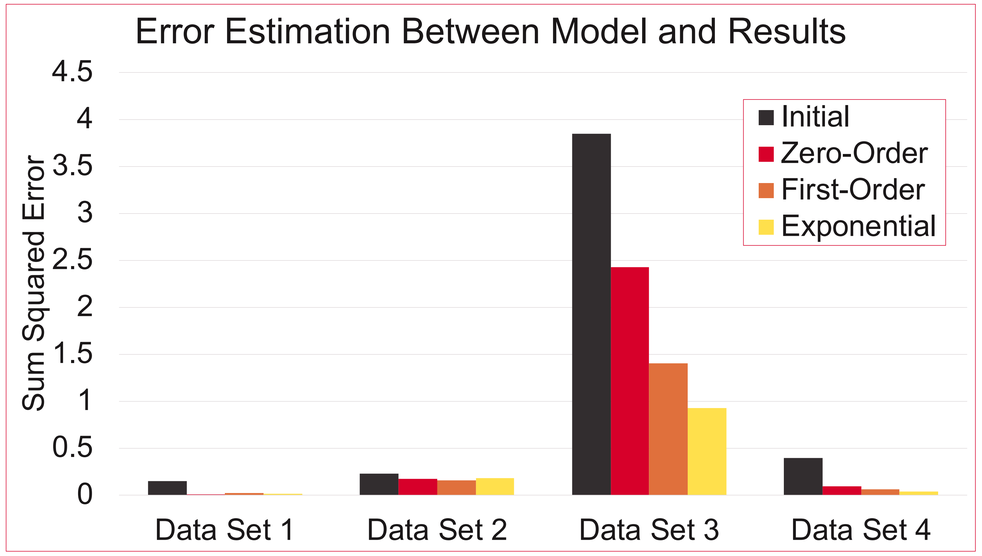


Data Set 3: Moderate Light, Low Nitrogen



Data Set 4: High Light, Low Nitrogen





Conclusions

- By introducing a degradation term, the error in the model was reduced overall.
- Consideration of the light intensity improved results when compared to a normal first-order degradation term.

Future Work

- Improved performance in the model may come from more understanding of the relationship between light and nitrogen stresses or by considering the effect of degradation on other metabolic systems.

Acknowledgements

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